



Enterprise Drone Operations & Inspection-Analytics Platform

A multi-tenant, serverless AWS platform that runs the full drone-inspection lifecycle for energy infrastructure — order, schedule, fly, ingest imagery, detect defects, and deliver geolocated analytics.

INDUSTRY

ENGAGEMENT

DELIVERY MODEL

PLATFORM

**Drone Services ·
Energy**

**~15-week program
(14 sprints)**

Fixed-scope Agile

**Serverless AWS +
iOS**

Confidentiality. Sanitized portfolio case study. The client and customer identities, product name, proprietary business rules, and internal source are withheld under NDA. Screenshots are from a demo/test environment with logos, order IDs, site names, and geolocation removed; outcome figures are representative and anonymized.

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AT A GLANCE

CHALLENGE

A drone-services operator ran inspections on email, spreadsheets, and a Heroku interim app — orders untracked, inspections non-standard, and gigabytes of flight imagery never turned into insight.

SOLUTION

A serverless, multi-tenant AWS platform for the full order → schedule → fly → ingest → detect → report lifecycle, with an asynchronous imagery pipeline and enterprise integrations.

RESULT

A scalable platform serving a dedicated anchor tenant plus shared-SaaS customers — digitized ordering, automated imagery-to-defect analytics, and executive cost/time-saved reporting.

~15 wk

Fixed-scope program delivered across 14 agile sprints

Order→Insight 2 tenancy models

Entire drone-inspection lifecycle digitized end-to-end

Dedicated-VPC anchor tenant + shared-SaaS customers

3 verticals

Solar, wind, and oil & gas inspections

CHALLENGE

Drone inspections stuck on email and spreadsheets

An enterprise drone-services company inspected energy infrastructure — solar, wind, oil & gas — for large industrial customers, with a global energy operator as its anchor customer adopting drone inspection at scale. The operation worked, but the tooling didn't keep up:

- **Orders lived in email.** Estimates, POs, and requests were untracked, with no online funnel from request to completed inspection.
- **No standardization.** Every inspection was scheduled, flown, processed, and reported differently — so cost, quality, and turnaround varied.
- **Imagery didn't become insight.** Raw flight logs and aerial imagery piled up, but turning them into geolocated defect maps was manual and slow.
- **No visibility.** Neither operator nor customers could see inspection KPIs, cost/time savings vs. manual inspection, or an audit trail.
- **Had to scale beyond one customer.** Serve the anchor customer as an isolated tenant *and* resell as multi-tenant SaaS.

SOLUTION

A serverless, multi-tenant drone platform

We designed and built a cloud-native platform on AWS that digitizes the entire order-to-insight lifecycle of a drone inspection — so a request becomes a geolocated, analyzed, auditable report, with heavy imagery processing pushed to an asynchronous pipeline that scales independently of the interactive app.

The delivery approach

- **Online ordering & workflow:** tracked estimates/orders with PO upload and review screens, a workflow engine moving each order request → scheduled → flown → processed → delivered, and email/in-app notifications.
- **Scheduling & a native flight app:** assign pilots, analysts, and drones; a native iOS "Ground Control" app (DJI SDK) captures flight logs and imagery in the field, with in-portal flight-log replay and telemetry.
- **Imagery pipeline & defect analytics:** raw captures land in S3 and flow through a Step Functions–orchestrated ETL (EMR/ECS/Lambda) into orthomosaics and geolocated, classified defect maps on ArcGIS, with inspection analytics and downloadable reports.
- **Executive analytics:** orders/flights over time, order mix by vertical, a submitted→completed funnel, and cost/time saved vs. manual inspection.
- **Multi-tenancy & integrations:** a dedicated VPC-isolated tenant for the anchor customer plus shared-SaaS tenancy; integrated with Salesforce (orders/catalog), SAP/Ariba (assets), and NetSuite (finance).

Architecture at a glance

```
Web (React) + iOS 'Ground Control' (DJI SDK)
|
Serverless API (API Gateway + Lambda · Node.js · Serverless Framework)
├─ Operational data: RDS PostgreSQL · DynamoDB · ElastiCache
├─ Integrations: Salesforce · SAP/Ariba · NetSuite
├─ Imagery pipeline: S3 (flight logs/imagery) → Step Functions
│   └─ ETL (EMR/ECS/Lambda) → processed inspections
│   └─ Elasticsearch · ArcGIS · Tableau
Multi-tenant: dedicated-VPC tenant + shared SaaS · Cognito · KMS · IAM · CI/CD
```

Technology

React Redux Redux-Saga Node.js Serverless Framework AWS Lambda API Gateway S3

CloudFront RDS (PostgreSQL) DynamoDB ElastiCache Step Functions EMR ECS Elasticsearch

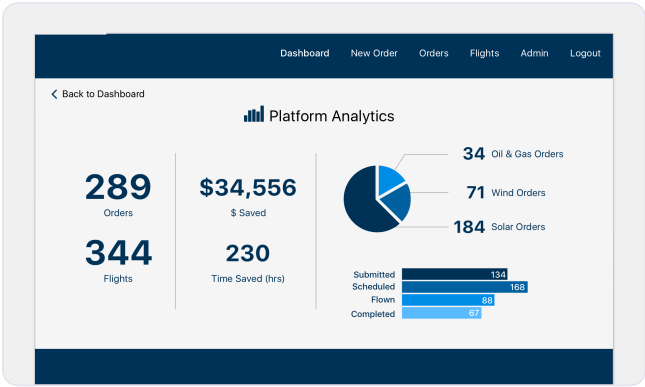
Cognito KMS SNS VPC CodePipeline iOS · DJI SDK Salesforce SAP / Ariba NetSuite

ArcGIS Tableau

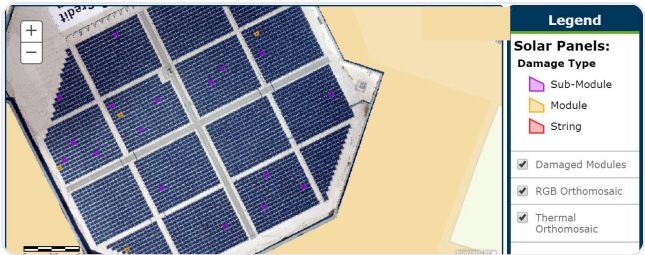
THE PRODUCT

From flight to geolocated insight

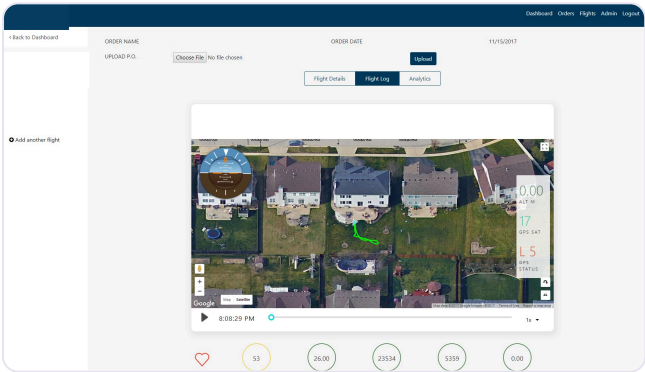
Demo/test environment. Logos, order IDs, site names, and geolocation redacted for NDA.



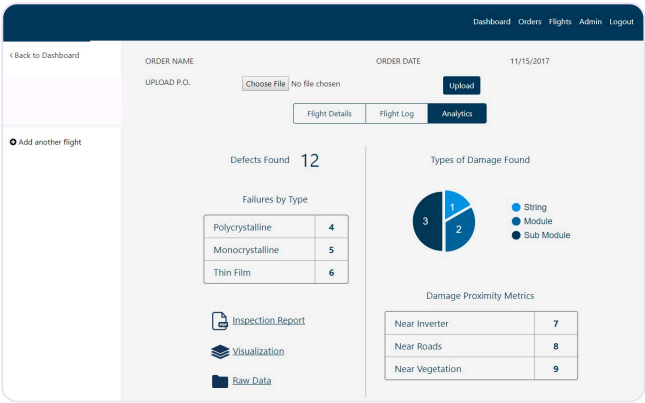
Platform analytics — orders and flights, order mix by vertical, and cost/time saved vs. manual inspection.



Geolocated defect map — a drone orthomosaic with classified damage points on an interactive GIS map.



Flight-log replay — the flown path on an aerial map with an attitude indicator and synchronized telemetry.



Inspection analytics — defects by type, damage classification, and proximity metrics.

RESULT**A scalable platform, off email and into the cloud****Email → Online**

Ordering moved off email/spreadsheets into a tracked online funnel

Imagery → GIS

Raw flights auto-processed into geolocated defect maps

Cost & time saved

Drone-vs-manual savings surfaced to operators & customers

- **One digital lifecycle.** Ordering, scheduling, flight capture, imagery processing, defect analytics, and reporting became a single tracked flow instead of email, spreadsheets, and manual hand-offs.
- **Imagery became insight — automatically.** An asynchronous pipeline turns raw drone flights into orthomosaics and geolocated, classified defect maps, with downloadable inspection reports.
- **Built to scale commercially.** A dedicated, VPC-isolated tenant for the anchor customer plus shared-SaaS tenancy — the same platform, resold across the energy industry.
- **Executive visibility.** Platform analytics quantify orders, flights, order mix, and cost/time saved vs. manual inspection — the KPIs the business was previously blind to.
- **Enterprise-integrated & serverless.** Salesforce, SAP/Ariba, and NetSuite integrations on a fully serverless AWS stack with CI/CD, delivered on a fixed-scope agile schedule.

Note on figures. Business-outcome figures shown in the product screenshots are demo/sample data. Specific client results, dataset schemas, credentials, site names, and identifying details are excluded under NDA; outcomes here are described qualitatively and anonymized.